



2T-Si/Sb₂S₃ tandem solar cell: Impedance spectroscopic and electrical circuit analysis

Atish Kumar Sharma^{a,b}, Ritesh Kumar Chourasia^{a,*}, Nitesh K. Chourasia^{c,*}, Keyur Sangani^{d,*}

^a Post-Graduate Department of Physics, Samastipur College (A Constituent Unit of L.N.M.U., Darbhanga-846004, Bihar, India), Samastipur 848134, Bihar, India

^b University Department of Physics, L.N.M.U., Darbhanga 846004, Bihar, India

^c Department of Physics, School of Energy Technology, Pandit Deendayal Energy University, Gandhinagar 382426, Gujarat, India

^d Institute of Technology, Nirma University, Ahmedabad, -382481, India

ARTICLE INFO

Keywords:

Capacitance-frequency
Mott-Schottky
Capacitance-voltage
Nyquist plots
EIS
Current matching condition

ABSTRACT

In the advancement toward sustainable energy solutions, solar power plays a pivotal role among renewable energy technologies. However, conventional single-junction solar cells are constrained by their intrinsic narrow bandgap, resulting in substantial energy losses specifically, thermalization losses from high-energy photons and transmission losses from low-energy photons. To address these limitations, tandem solar cells offer a promising alternative by enabling the absorption of a wider portion of the solar spectrum, thereby significantly mitigating both thermal and absorption-related losses. This study presents a comprehensive analysis of a two-terminal (2 T) tandem solar cell architecture utilizing impedance spectroscopy and equivalent circuit modelling. Using SCAPS-1d, the standalone top cell (Sb₂S₃) and bottom cell (Si) cells are demonstrated. The EIS analyser is used for fitting the equivalent circuit corresponding to the frequency-dependent impedance data for the top and bottom cells. The characterization of top and bottom cells is done exclusively based on the capacitance-frequency (C–F), capacitance-voltage (C–V), Mott-Schottky (M–S), and Nyquist plots. The impedance spectroscopy data of both cells confirms their suitability as top and bottom cells for the 2 T tandem design.

1. Introduction

The global demand for renewable energy has driven significant advancements in photovoltaic technologies, with tandem solar cells emerging as a leading strategy to achieve efficiencies beyond the Shockley-Queisser limit of single-junction devices (29.4% for crystalline silicon) [1,2]. Tandem solar cells stack two or more subcells with complementary bandgaps to capture a broader range of the solar spectrum, thereby improving overall energy conversion efficiency [3]. Among the diverse tandem solar cell configurations, the combination of antimony trisulfide (Sb₂S₃) as the top cell and silicon (Si) as the bottom cell offers a compelling synergy of high performance, cost-effectiveness, and environmental sustainability. This study investigates the potential of Sb₂S₃/Si tandem solar cells, with a focus on the optoelectronic properties of the constituent materials, their compatibility in tandem architectures, and the key challenges and opportunities associated with improving their overall performance.

Antimony trisulfide (Sb₂S₃) is a binary chalcogenide semiconductor with a bandgap of approximately 1.7–1.8 eV, making it well-suited for

absorbing high-energy photons in the visible spectrum. Its high absorption coefficient ($>10^5 \text{ cm}^{-1}$) allows efficient light capture in thin films, reducing material usage and enabling integration into tandem structures [4–7]. Orthorhombic crystal structure of Sb₂S₃, characterized by one-dimensional (Sb₄S₆)_n ribbons, imparts anisotropic electrical and optical properties that require careful optimization but offer unique opportunities for charge transport engineering [8–11]. The earth-abundant, non-toxic nature and exceptional environmental stability of Sb₂S₃ render it a promising alternative to conventional photovoltaic materials such as cadmium telluride (CdTe) and indium-based compounds, which are often limited by toxicity and resource scarcity. Recent advancements in deposition techniques, such as chemical bath deposition (CBD) and atomic layer deposition (ALD), have significantly improved the structural and morphological quality of Sb₂S₃ thin films. However, the power conversion efficiency of single-junction Sb₂S₃ devices remains constrained to around 10%, largely due to recombination losses and defect-related limitations. [12]. Silicon, with its indirect bandgap of $\sim 1.1 \text{ eV}$, is the foundation of the photovoltaic industry, dominating over 90% of the global market share due to its reliability,

* Corresponding authors.

E-mail addresses: riteshphysics@gmail.com (R.K. Chourasia), niteshphysics@gmail.com (N.K. Chourasia), sanganikeyur3670@gmail.com (K. Sangani).

<https://doi.org/10.1016/j.jelechem.2026.119895>

Received 14 December 2025; Received in revised form 20 January 2026; Accepted 28 January 2026

Available online 30 January 2026

1572-6657/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1

The simulation parameters of the Sb_2S_3 top cell of the proposed tandem solar cell configuration [4].

Layers parameter	FTO [34,35]	TiO_2 [29–31]	Sb_2S_3 [7,27,28]	V_2O_5 [32,33]
Thickness (μm)	0.05	0.05	0.427	0.02
Bandgap (eV)	3.8	3.1	1.71	2.8
Electron Affinity (eV)	4.1	4.2	3.7	2.6
Density of States Conduction Band ($/\text{cm}^3$)	2.2E18	2.2E18	5E19	9.2E19
Density of States Valence Band ($/\text{cm}^3$)	1.8E18	1.8E18	1E20	5E20
Relative permittivity	9	9	5	8
Electron mobility (cm^2/Vs)	100	0.8	9.8	6
Hole mobility (cm^2/Vs)	25	0.2	0.2	1
Acceptor density ($/\text{cm}^3$)	0	0	1E16	1E16
Donor density ($/\text{cm}^3$)	1E18	1E17	0	0
Defect density ($/\text{cm}^3$)			1.5E14	1E14
Type of Defect			Acceptor (0/-)	Neutral
Capture cross section of electrons (cm^2)			3E-15	1E-15
Capture cross section of holes (cm^2)			3E-15	1E-15

Table 2

The simulation parameters of the Si-bottom cell of the proposed tandem solar cell [4].

Layers parameter	n + -Si	p-Si	p + -Si
Thickness (μm)	0.74	100	2.1
Bandgap (eV)	1.12	1.12	1.12
Electron Affinity (eV)	4.05	4.05	4.05
Density of States Conduction Band ($/\text{cm}^3$)	1E19	1E19	1E19
Density of States Valence Band ($/\text{cm}^3$)	1E19	1E19	1E19
Relative permittivity	11.9	11.9	11.9
Electron mobility (cm^2/Vs)	60.8	1099	193.4
Hole mobility (cm^2/Vs)	151.9	426.9	45.1
Acceptor density ($/\text{cm}^3$)	0	1E16	2E20
Donor density ($/\text{cm}^3$)	2E20	0	0

scalability, and well-established manufacturing infrastructure [13,14].

Its bandgap is well-suited for the absorption of lower-energy photons in the near-infrared region, effectively complementing the higher-energy photon absorption of Sb_2S_3 in a tandem configuration. Additionally, silicon exhibits high carrier mobilities, approximately $1400 \text{ cm}^2/\text{V}\cdot\text{s}$ for electrons and $450 \text{ cm}^2/\text{V}\cdot\text{s}$ for holes in monocrystalline form as well as long carrier diffusion lengths, both of which contribute to efficient charge carrier collection. [15]. Advanced silicon cell architectures, such as passivated emitter and rear cell (PERC) or heterojunction with intrinsic thin layer (HIT), have pushed single-junction efficiencies above 26%, making silicon a robust foundation for tandem devices. Its abundance, non-toxicity, and operational lifetime exceeding 25 years further enhance its appeal for large-scale deployment [16,17].

The $\text{Sb}_2\text{S}_3/\text{Si}$ tandem configuration leverages the complementary bandgaps of the two materials to achieve efficient spectral splitting. The wide-bandgap Sb_2S_3 top subcell absorbs photons with energies above $\sim 1.7 \text{ eV}$, while allowing longer-wavelength photons to pass through to

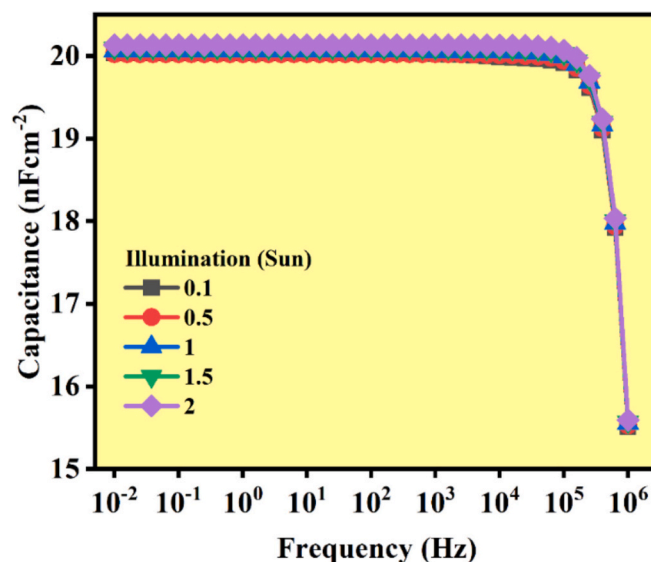


Fig. 2. C-f characteristics of the top cell under various illumination levels from 0.1Sun to 2Sun.

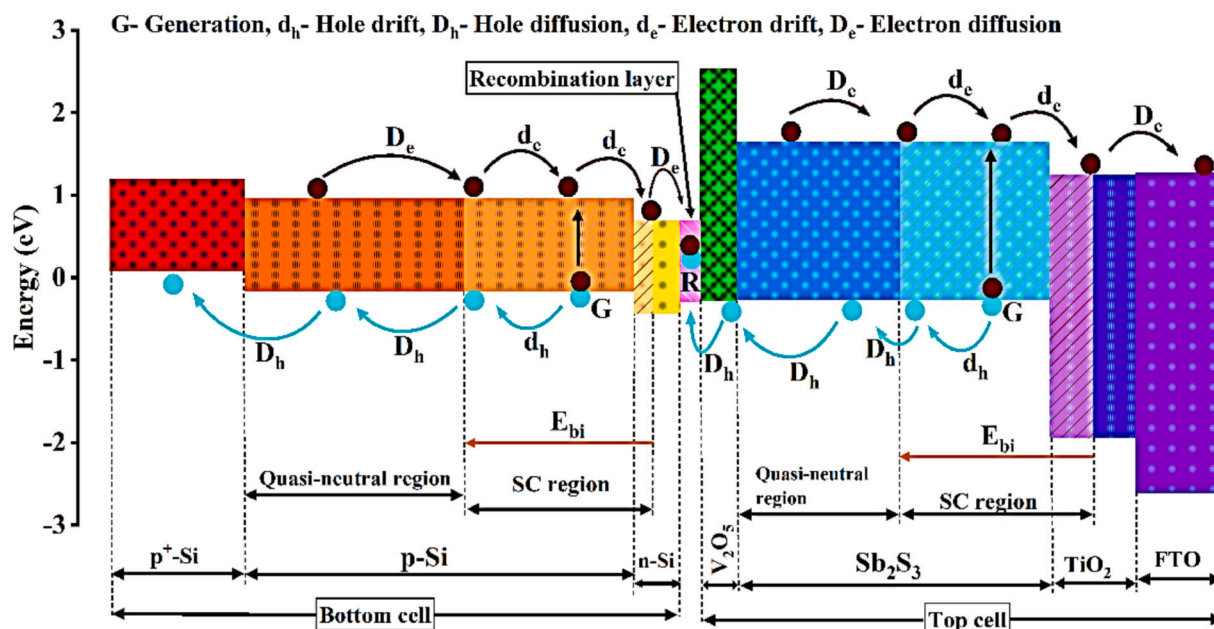


Fig. 1. Energy level block diagram of the top and bottom cells in a tandem configuration with charge transportation mechanism through various layers.

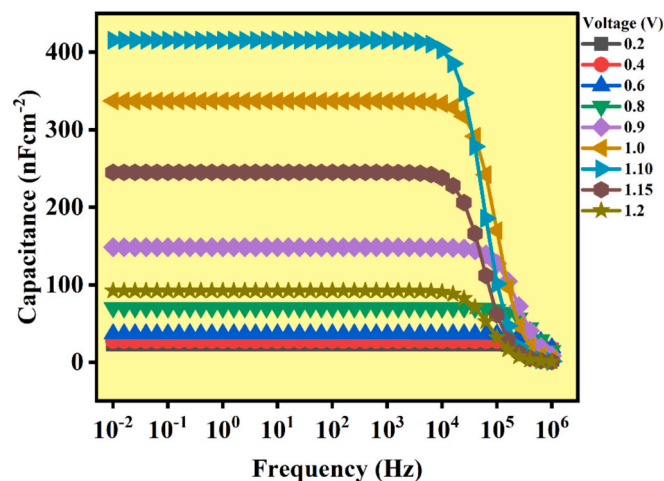


Fig. 3. C-f characteristics of the top cell under biasing voltages from 0.2 V to 1.2 V.

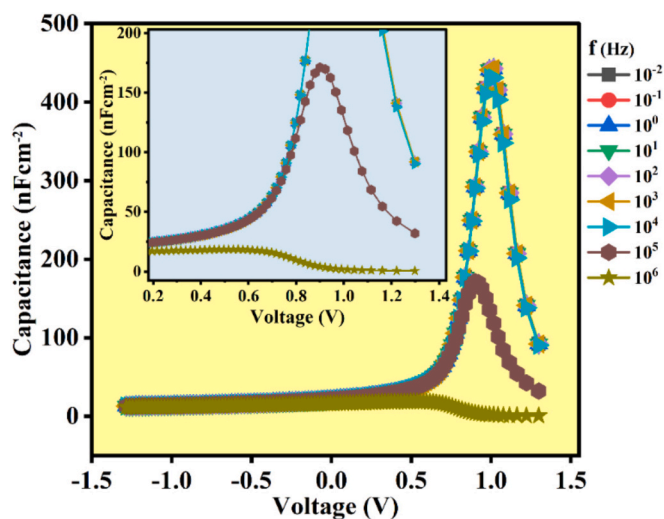


Fig. 4. C-V characteristics of the top Sb_2S_3 cell of the proposed tandem configuration under different frequencies.

the Si-bottom cell. This combination theoretically enables efficiencies exceeding 30%, surpassing the limits of single-junction silicon cells. The use of a transparent conductive interlayer, such as indium tin oxide (ITO), facilitates electrical and optical coupling between the cells, ensuring efficient charge transfer and minimal parasitic absorption. Both monolithic (2-terminal) and mechanically stacked (4-terminal) designs are viable, with the former requiring precise current matching and the latter offering greater flexibility in subcell optimization [4,18–20]. Fabricating high-efficiency $\text{Sb}_2\text{S}_3/\text{Si}$ tandem solar cells requires addressing several challenges. For the Sb_2S_3 top subcell, optimizing film quality to reduce recombination losses and control crystal orientation is critical, as its anisotropic structure can impede charge transport if not properly managed [21,22]. Deposition techniques must balance uniformity, scalability, and cost, with methods like ALD and spin-coating showing promise. For the silicon bottom subcell, advanced passivation techniques and light-trapping strategies are essential to maximize efficiency. In monolithic tandems, achieving current matching between the subcells is a significant hurdle, necessitating precise control over the thickness and optical properties of the Sb_2S_3 layer. The interlayer between subcells must minimize parasitic absorption and ensure low-resistance electrical connectivity. Additionally, integrating

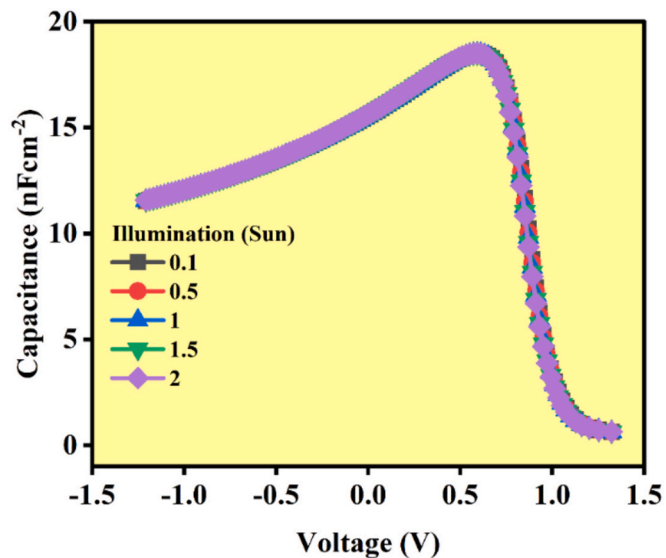


Fig. 5. Illumination-dependent (from 0.1 Sun to 2 Sun) C-V characterization of the top cell.

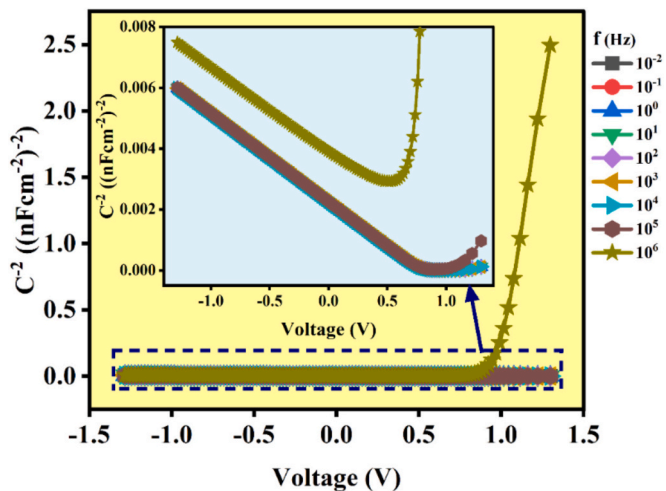


Fig. 6. Frequency-dependent (from 10^{-2} Hz to 10^6 Hz) Mott-Schottky plot of the top cell (inset showing the enlarged scale of the vertical axis to identify intercept on the voltage axis).

the relatively immature Sb_2S_3 technology with the mature silicon platform requires careful process compatibility to avoid thermal or chemical degradation [23]. Furthermore, to identify the process of governing the solar cell mechanism under various conditions, we employed the impedance spectroscopy analysis. This is one of the most significant characterization techniques for electrochemical systems and solid state devices in a broader sense of materials, such as organic, inorganic, as well as biological systems [24]. The impedance spectroscopy, which is the reciprocal of the admittance spectroscopy, involves the frequency-dependent capacitance measurements [25]. This method provides insight into the physical and chemical processes of devices at their interfaces and in the bulk, doping concentration, trap state, and trap density behavior.

In the present communication, we first presented the impedance spectroscopy of the top $\text{FTO}/\text{TiO}_2/\text{Sb}_2\text{S}_3/\text{V}_2\text{O}_5$ cell. The top cell is simulated for the C-F characteristics under various illumination levels, applied bias voltages, and various working temperatures. Also, the electrical equivalent circuits of the top cell are drawn based on the

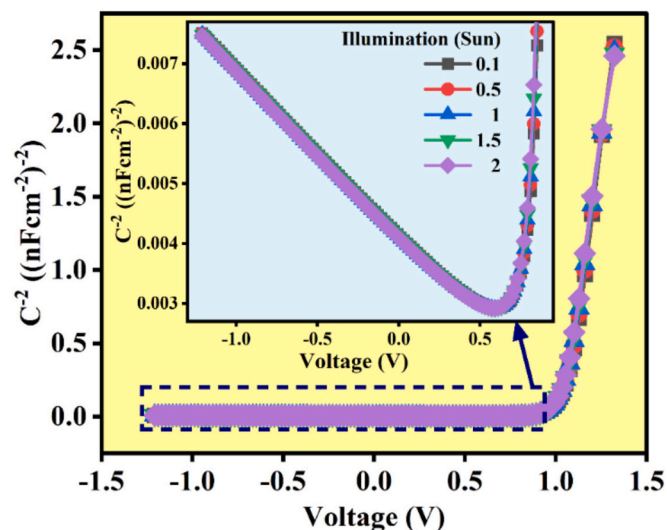


Fig. 7. Illumination-dependent (from 0.1Sun to 2Sun) Mott-Schottky plot of the top cell, and the inset shows an enlarged vertical axis.

impedance spectroscopy at various voltages. Further, the impedance spectroscopy of the Si-bottom cell is performed under various illumination levels, biasing voltages, and at various temperatures. The structure of this work is organized as follows: Section 1 presents a detailed introduction to the study. Section 2 outlines the device configurations and describes the methodologies employed. Section 3 elaborates on the theoretical framework underlying the investigation. Section 4 discusses the results derived from the study, with Section 4.1 focusing on the analysis of the Sb_2S_3 top cell and Section 4.2 examining the performance of the silicon bottom cell. Finally, Section 5 concludes the study with a summary of key findings and significant remarks.

2. Device configuration and simulation methodology

The presented top cell and bottom cell, as well as the 2 T-tandem configuration solar cells, are modeled and simulated using SCAPS-1D solar cell simulation software. As SCAPS 1D requires material parameters for device modelling therefore, we have adopted the necessary physical parameters that are close to the reported values in experimental as well as theoretical literatures. Also, the SCAPS has default models for some of the parameters such as absorption models, defect models, etc. [26]. Therefore, bandgap, electron affinity, mobility, and relative permittivity are taken as reported in the literatures for the materials as Sb_2S_3 [7,27,28], TiO_2 [29–31], V_2O_5 [32,33], FTO [34,35] for top

subcell. Further, for the Si bottom subcell materials also the material parameters are adopted as reported in the literatures [36,37]. The parameters utilized in the current simulation of top and bottom cells are summarized in Tables 1 and 2, respectively [4]. The energy level block diagram of all the layers of the top and bottom cells in the tandem solar cell configuration is shown in Fig. 1.

3. Theoretical discussion

Since the standalone cells in a tandem solar cell are stacked in a descending order of their bandgap, they absorb the photons of the incident solar spectrum according to their bandgaps. In SCAPS-1D, the absorption of the incident solar spectrum is governed by the E_g -sqrt absorption model, which depends on the bandgap as follows:

$$\alpha(h\nu) = \left(\alpha_0 + \frac{\beta_0 E_g}{h\nu} \right) \left(\frac{h\nu}{E_g} - 1 \right)^{1/2} \quad (1)$$

where, α_0 and β_0 are the material constants with dimensions similar to the absorption coefficient α . E_g and $h\nu$ are the energy bandgap and photon energy. The absorption coefficient $\alpha(h\nu)$ in eq. 1 becomes zero when the photon energy is less than the bandgap [38].

Impedance spectroscopy is a challenging technique to analyse the electrical behaviours of electrochemical systems [39]. The solar cells have solid electrodes and layers between them such as absorber, ETL, and HTL serve as electrolytes [40]. Therefore, electrochemical impedance spectroscopy (EIS) is employed to investigate the impedimetric and capacitive characteristics of the proposed solar cell architecture. The relation between capacitance and frequency is given by the reactive impedance of the electric circuit X_c as in eq. 1.

$$X_c = \frac{1}{jC\omega} \quad (2)$$

where, $j = \sqrt{-1}$, C is the capacitance and the ω is the angular frequency of the signal. The photo-capacitance (C_{ph}) as a function of the generation rate (G) by illumination and mobility is given by the relation as in eq. 2 [41].

$$C_{ph} = \frac{qG}{2(\mu_p + \mu_n)} \left(\frac{d^3}{(V_{bi} - V)^2} \right) \quad (3)$$

where, μ_p and μ_n are the hole and electron mobilities, and the V_{bi} is the built-in voltage. The D. C. biasing voltage applied to the solar cell is V . Since the mobilities of the material for the solar cell are fixed as opted from the literature. Thus, for a particular thickness, the capacitance directly varies with the generation rate of the illumination. Also, the

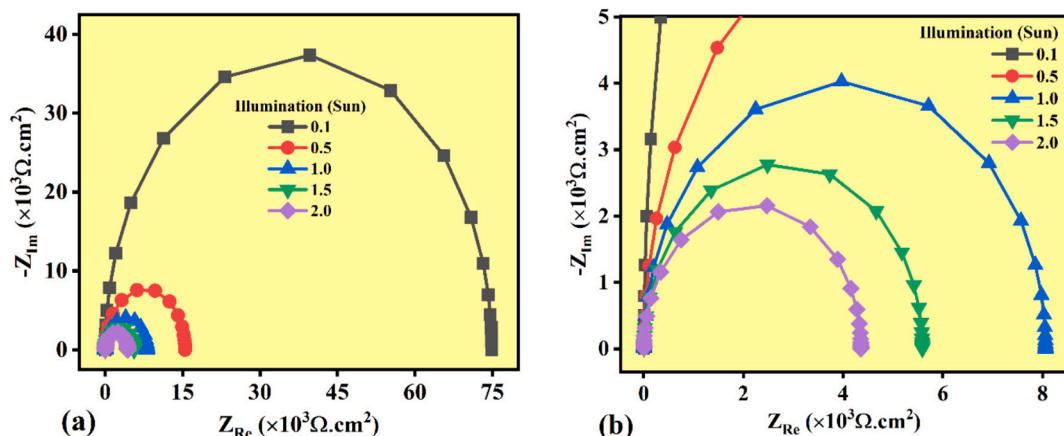


Fig. 8. (a) Illumination-dependent Nyquist plots of the top cells, and (b) highlight the plots for higher illuminations.

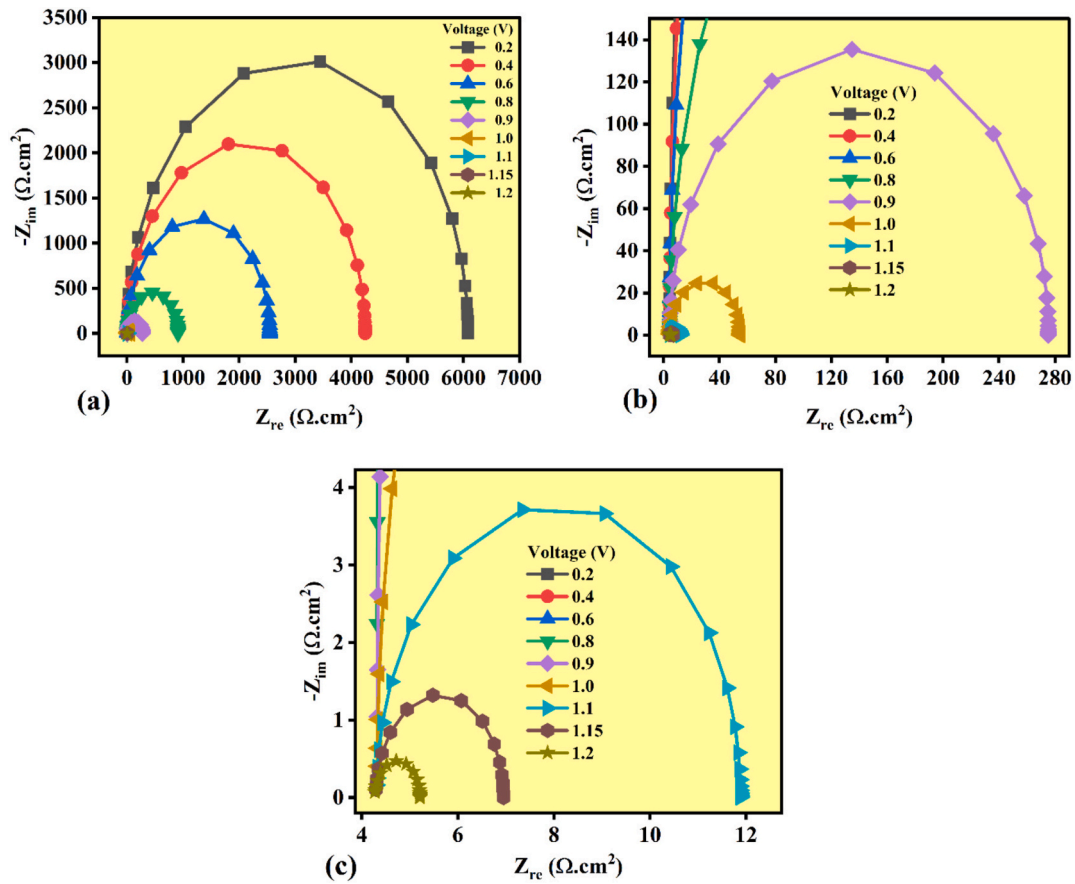


Fig. 9. (a) Biasing-dependent Nyquist plots for the top cell of the proposed tandem solar cell, (b) and (c) show the enlarged view of plot (a).

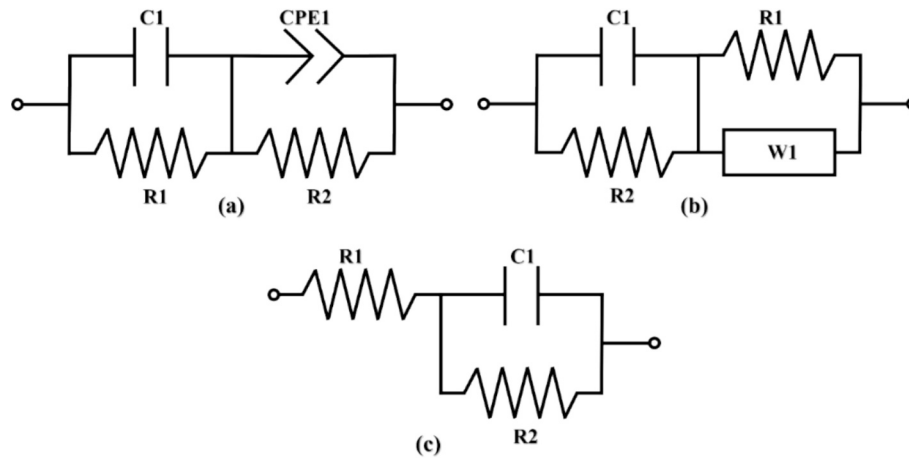


Fig. 10. Electrical equivalent circuits for the top cell for different biasing voltages as (a) 0.2, 0.4, and 0.9 V, (b) 0.6, 0.8, and 1.0 V, (c) 1.1, 1.15, and 1.2 V.

built-in voltage and the defect density are related to the capacitance by the Mott-Schottky relation [42] given by eq. 3.

$$\frac{1}{C^2} = \frac{-2}{\epsilon \epsilon_0 q A^2} \left(\frac{V - V_{bi}}{N_t} \right) \quad (4)$$

where, C is the total capacitance, ϵ is the absolute permittivity, ϵ_0 is the permittivity of the vacuum and N_t is the density of defects. The area of the cell is A which is 1 cm^2 for our simulation, thus the Mott-Schottky relation can be given as in Eq. 4.

$$\frac{1}{C^2} = \frac{-2}{\epsilon \epsilon_0 q} \left(\frac{V - V_{bi}}{N_t} \right) \quad (5)$$

The total capacitance offered by the solar cell is the sum of the dark capacitance and the photo capacitance. The dark capacitance is associated with the geometry and depletion of the cell, whereas the generation rate contributes to the photo capacitance. Thus, the total capacitance can be given as in eq. 5 in the presence of illumination [41].

$$C_{Tot} = C_{ph} + C_{dl} \quad (6)$$

where, C_{dl} is depletion capacitance, and the total capacitance is C_{Tot} .

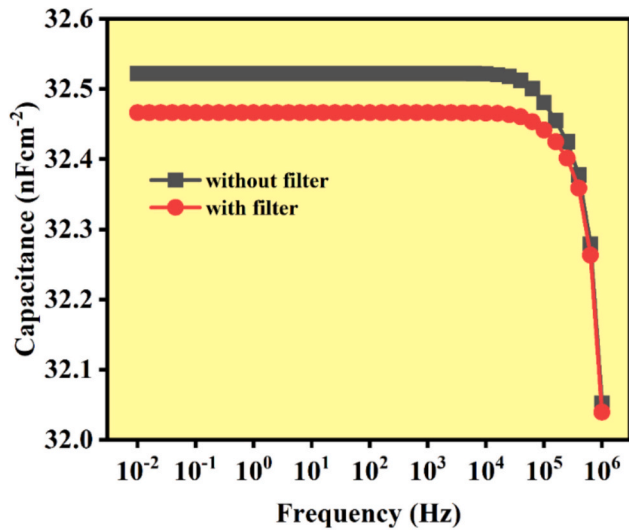


Fig. 11. C-f characteristics of the bottom cell without filter (under AM 1.5 G 1Sun spectrum) and with filter (under AM_1.5G 1Sun filtered after top cell).

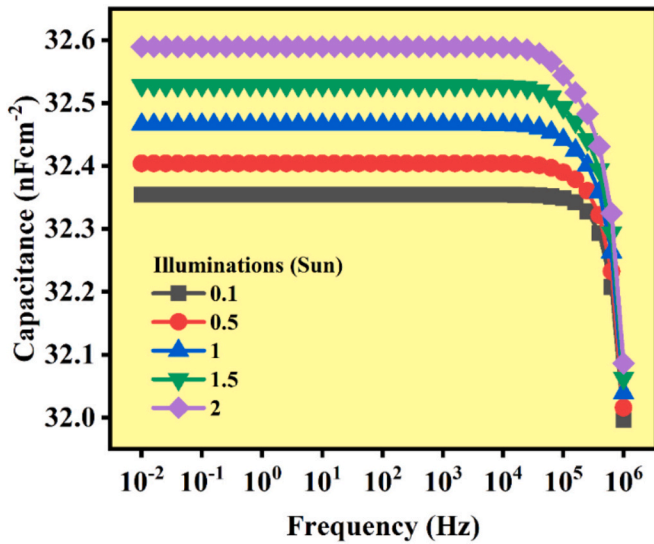


Fig. 12. Illumination-dependent C—F characteristics of the proposed Si bottom cell of the tandem configuration.

Also, the depletion capacitance as a function of doping densities and the voltages can be expressed as in Eq. 6.

$$C_{dl} = A \left[\frac{N_A N_D q \epsilon}{2(V_{bi} - V)(N_A + N_D)} \right]^{1/2} \quad (7)$$

Where, N_A and N_D are the acceptor and donor densities.

For the subsequent solar cells in the stack of the tandem solar cell configuration, the filtered spectrum $S(\lambda)$ is estimated using the relation given as in eq. 7 [43].

$$S(\lambda) = S_o(\lambda) e^{\sum_{i=1}^{i=n} (-\alpha_i(\lambda) d_i)} \quad (8)$$

where, $S_o(\lambda)$ is the wavelength (λ) dependent incident solar spectrum AM1.5G 1Sun spectrum, $\alpha(\lambda)$ is the wavelength-dependent absorption coefficient of the materials of the solar cell, d is the thickness of the layers of the materials used in the solar cell. The index i is denotes the layers for $i = 1$ to n , n being the number of layers through which the incident spectrum travels. For our simulation $i = 1$ to n is $i = 1$ to 4 where 1,2,3, and 4 represents FTO, TiO_2 , Sb_2S_3 , and V_2O_5 respectively.

4. Results and discussion

In the present research, we have explored the electrical properties of the two sub-cells appropriate for the 2 T tandem solar cell application. This exploration is performed utilizing a well-established modelling and simulation software, SCAPS-1D [38]. The two sub-cells (FTO/ TiO_2 / Sb_2S_3 / V_2O_5 top sub-cell and n-Si/p-Si/p⁺-Si bottom sub-cell) are simulated independently, and the corresponding results are presented. The top cell is simulated under the standard AM 1.5G 1 sun spectrum radiation, while for the simulation of the bottom cell, a filtered spectrum is applied. Initially, the equivalent electrical circuits for the individual cells were modeled using frequency-dependent impedance data. Subsequently, further analysis was conducted through capacitance–frequency (C–F) and capacitance–voltage (C–V) characterizations to gain deeper insights into the device behavior.

4.1. Impedance spectroscopy of the top cell

To investigate the top cell using impedance spectroscopy, its C–F characteristics were simulated under various illumination conditions, as shown in Fig. 2. The results indicate that the capacitance remains approximately constant at around 20 nFcm^{−2}, exhibiting negligible dependence on frequency up to 10⁵ Hz. However, for the frequency above 10⁵ Hz, the capacitance is reduced to 15.59 nFcm^{−2}. Since, in the high-frequency input signal, the reactive impedance increases, therefore the capacitance decreases, which can be verified from eq. 2. Additionally, the illumination intensity varying in the range of 0.1 Sun to 2 Sun affected the capacitance minutely. This effect becomes trivial because of the increase in mobility and conductivity as well as generation rate with the sunlight in the specified range of intensity, which follows eq. 3.

Further, the C–F characteristic of the top sub-cell is also observed under different biasing levels from 0.2 V to 1.2 V, as plotted in Fig. 3. In the lower frequency region nearly less than 10⁵ Hz, the capacitance is affected by the biasing levels significantly. In this region, the capacitance increases with the biasing. The observed increase in capacitance is attributed to the accumulation of photogenerated charge carriers under the applied bias. This enhancement continues up to the open-circuit voltage (Voc), beyond which the capacitance behavior changes due to a sharp rise in current following breakdown at Voc. Thus, the capacitance is highest for the biasing level near the Voc of the top cell, and for the biasing other than the Voc, the capacitance is minimal. However, for the highest frequency, the capacitance is lowest for all the biasing levels. Moreover, in the high-frequency region, the capacitance is lowest for all the biasing levels and shows consistency with the biasing. This shows the independence of the capacitance over the high frequency.

The frequency-dependent C–V characteristic is measured under the variation of 10^{−2} Hz and 10⁶ Hz and is displayed in Fig. 4. For all the frequencies, the capacitance is negligible and remains independent of the voltage till 0.5 V. The capacitance for lower frequency curves near Voc, i.e., around 1.159 V, increased exponentially to 450 nFcm^{−2} owing to the maximum recombination at Voc. Further, for voltages above the Voc, the capacitance decreases, which is in the diode region.

In addition to the frequency-dependent C–V characteristics, the illumination-dependent C–V behavior was also simulated, with illumination levels ranging from 0.1 Sun to 2 Sun, as shown in Fig. 5. For the voltage up to about 0.7 V, the capacitance increases almost linearly from 11 nFcm^{−2} to 18 nFcm^{−2}. For higher voltages, the capacitance falls exponentially with the applied voltage. Furthermore, the capacitance remains almost independent of illumination intensity. This behavior of the C–V characteristics is due to the reduction in the depletion region with the applied bias voltage. The contraction of the depletion region leads to a significant increase in the dark (or depletion) capacitance, as described in eqs. 3, 6, and 7. Consequently, the influence of the applied bias voltage on the capacitance is more pronounced than the reduction effect caused by illumination.

The defect density within the bulk material and at the interfaces

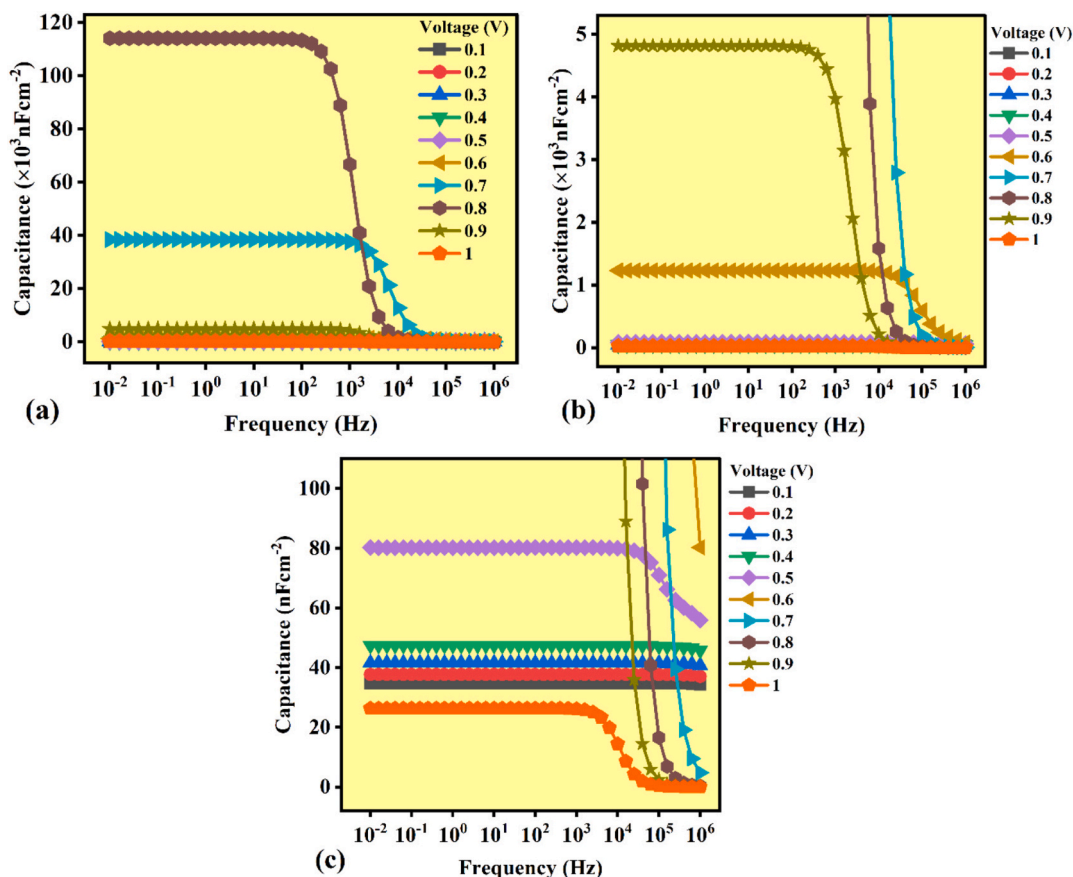


Fig. 13. (a) Voltage-dependent C–F characterization of bottom cells, (b), and (c) shows the enlarged plots of (a).

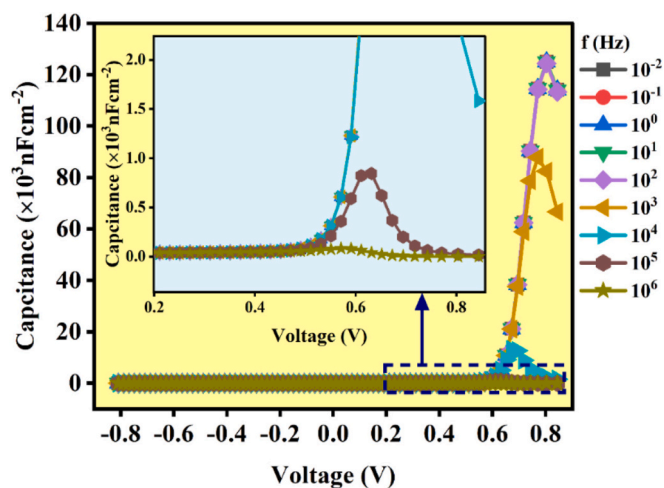


Fig. 14. Frequency-dependent C–V characteristics of the Si-bottom cell of the proposed tandem configuration.

between layers significantly influences the performance parameters of photovoltaic cells. To investigate the behavior of defect states in the proposed solar cell, the Mott–Schottky (MS) characteristics were analyzed, as illustrated in Fig. 6. For all the frequencies, the voltage-dependent C^{-2} value shows negligible variation. The inset of Fig. 6 represents the precise variation of the C^{-2} with the applied biasing voltage. The intercept on the voltage axis of the MS curve gives the built-in voltage, which is estimated by eqs. 3 to 5. Thus, from the plots in Fig. 6, it is clear that the built-in voltage is about 0.8 V for the top cell of

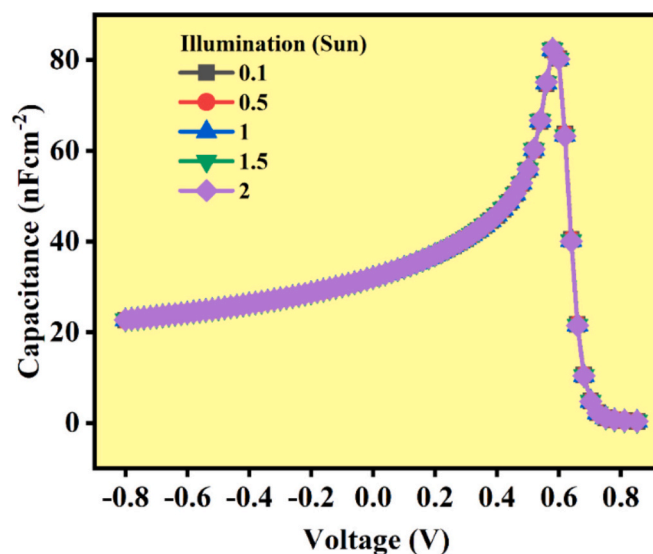


Fig. 15. Illumination-dependent C–V characteristics of the Si-bottom cell of the proposed tandem solar cell configuration.

the proposed tandem cell. Also, we observed that, MS plot is helpful in the lower frequency region. For the biasing levels up to near V_{oc} , the MS plots show invariance with all the frequencies; thereafter, it varies slightly for frequencies up to 10^5 Hz. But for the frequency of 10^6 Hz, the MS plot shows rapid increase with the biasing level above the V_{oc} .

Along with the frequency-dependent MS plot, the illumination-dependent MS plot is also presented as shown in Fig. 7. The MS plots

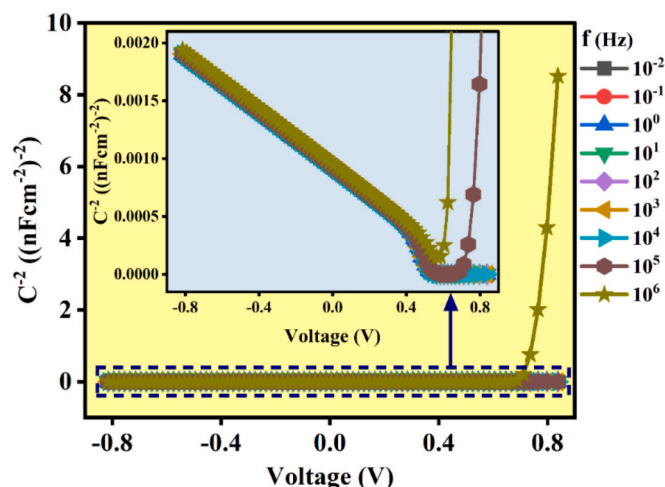


Fig. 16. Frequency-dependent Mott-Schottky plot of the Si-bottom cell of the proposed tandem solar cell configuration.

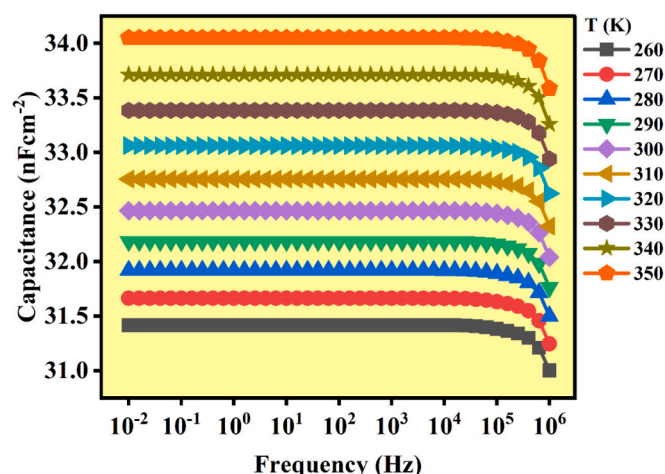


Fig. 17. Temperature-dependent C-F characterization of the Si-bottom cell of the proposed tandem solar cell configuration.

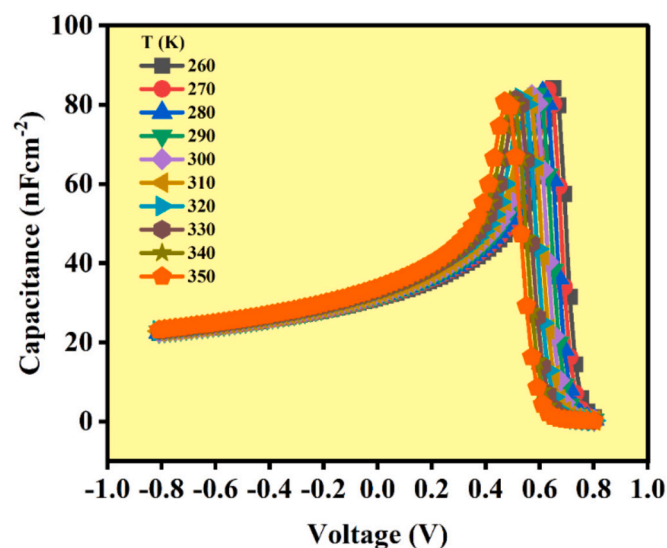


Fig. 18. Temperature-dependent C-V characterization of the Si-bottom cell of the proposed tandem solar cell configuration.

with all the illumination levels show similar variations at all the biasing levels. The inset of Fig. 7 shows the precise variation of the MS plots with biasing levels, which are observable. From the inset of Fig. 7, we observe the built-in voltage of about 0.6 V. Above the V_{oc} , for all the illumination levels, the MS plots increase rapidly with the applied biasing voltages.

Researchers have carried out few studies on impedance spectroscopy of the Sb_2S_3 based solar cell experimentally [44–46]. Their device configuration are somehow similar to our existing device architecture in present manuscript. They have used impedance spectroscopy to optimize source substrate proximity [46]. Some researchers used this technology to optimize back interface in Sb_2S_3 solar cell to reduce back interfacial recombination [45,46]. Thus, we have performed impedance spectroscopy studies for our device configurations (Top and Bottom cells) utilized in the tandem solar cell. Our findings related to frequency dependent impedance spectroscopy are closed to the recently reported outcomes [47].

The impedimetric behavior of the top cell at room temperature is observed under various illumination levels. For the examination of the impedimetric behavior of the top cell, the illumination-dependent Nyquist plots are intended in Fig. 8 (a & b). Fig. 8b presents a magnified view of Fig. 8a at a reduced scale to more clearly highlight the variations. The size of semicircles decreases with increasing illumination levels. As the illumination level increases, the generation rate increases. This increasing generation raises the carrier concentration in the cell, which further contributes to the conduction of the carriers. Consequently, the impedance of the cell decreases with the illumination levels, which is evidenced by the shortening of the radii of the semicircles.

Moreover, in Fig. 9 (a to c), the Nyquist plots are drawn with the variation of the biasing voltage levels in the range of 0.2 V to 1.2 V. The Nyquist plots are visible from 0.2 V to 0.8 V in Fig. 9a), from 0.9 V to 1.0 V in Fig. 9b), and from 1.1 V to 1.2 V in Fig. 9c). From the Nyquist plots, we discern that the radii of the semicircles are shrinking with the increase in the applied bias voltage. This shrinking of Nyquist plots is attributed to the increase in the recombination of the photogenerated carriers with the applied bias voltage.

The electrical equivalent circuits (EEC) of the top cell for different biasing levels are shown in Fig. 10. These EECs are obtained by fitting the C-F data (as availed from the SCAPS 1D C-F calculation) using the Electrochemical impedance analyser software (EIS Analyser software) [48]. The components R_1 , R_2 , and C_1 are the resistances and capacitances offered by the top cell layers and interfaces. CPE1 is the constant phase element, which corresponds to the non-ideal capacitive behavior of a solar cell at higher frequencies [49]. Further, W_1 represents the Warburg element, which corresponds to the impedance due to the diffusion mechanism at lower frequencies [50]. Thus, it can be observed that for the bias voltages 0.2, 0.4, and 0.9 V, the top cell shows non-ideal capacitive behavior. Further, at moderate biasing voltages such as 0.6, 0.8, and 1.0 V, the diffusion process prevails. Whereas for higher biasing levels such as 1.1, 1.15, and 1.2 V, both the non-ideal capacitance as well as the diffusion process vanish.

4.2. Impedance spectroscopy of bottom cell

Analogous to the study of the top cell based on impedance spectroscopy, the bottom cell is also studied on a similar basis. The C-F characteristics of the Si bottom cell are simulated under two (filtered and non-filtered) spectra as shown in Fig. 11. For the C-F characterization of the Si bottom cell, we applied the filtered spectrum, which was the leftover after the absorption in the top cell. Therefore, in Fig. 11, the C-F characterization is represented for both with and without filter conditions. As displayed in Fig. 11, the capacitance of the Si bottom cell is obtained to be slightly less with the filtered spectrum. This shows that the capacitance of the Si bottom cell is minutely affected by the filtered spectrum. Thus, the capacitance contribution by the Si bottom cell in the tandem structure will be affected slightly.

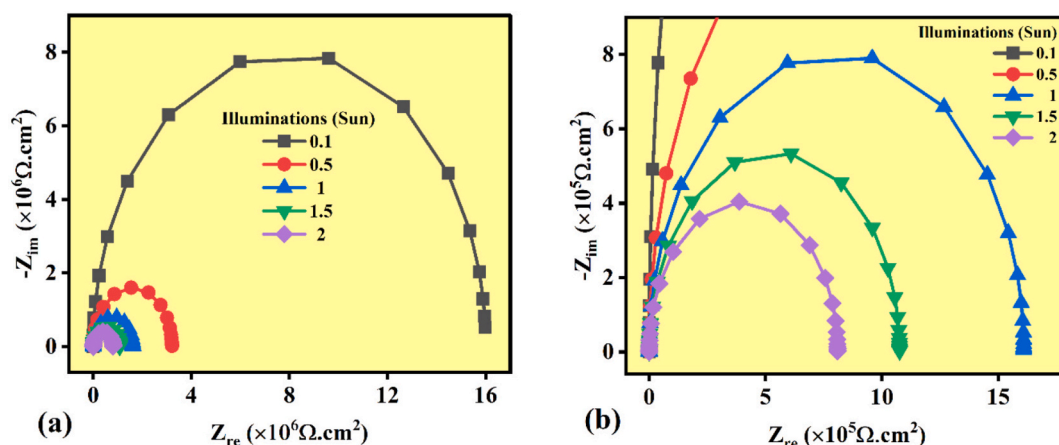


Fig. 19. (a) Illumination-dependent Nyquist plots for the bottom cell, and (b) highlight the plots for higher illumination levels.

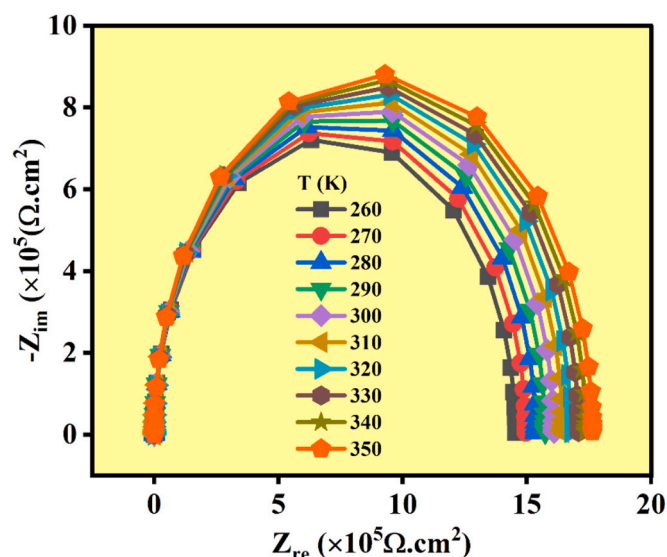


Fig. 20. Temperature-dependent Nyquist plots for the Si-bottom cell of the proposed tandem solar cell configuration.

Further, the effect of the illumination levels in the range of 0.1 Sun to 2 Sun on the C–F characteristics of the Si bottom cell is examined, as represented in Fig. 12. The illumination-dependent C–F characteristics of the Si bottom cell increased slightly from 32.3 nFcm^{-2} to 32.6 nFcm^{-2} . From eq. 3, we can see that photo capacitance depends linearly on the generation rate under illumination. Thus, the slight increase in capacitance of the Si bottom cell is due to the rise in photogenerated carriers in the device with increasing illumination levels.

Extending further with the spectroscopic study, the C–F characterization of the Si bottom cell is examined under various applied bias voltages as exhibited in Fig. 13 (a to c). The capacitance of the Si bottom cell is highest for the bias voltage of 0.8 V and becomes lowest at the minimum and maximum voltages. The voltage-dependent C–F characteristic shows that the capacitance is effective only for the lower frequency range below 10^5 Hz . Whereas the capacitance is mostly invariant in the high-frequency region above 10^6 Hz . This suggests that the charge carriers contributing to the capacitance are responsive mainly to the lower frequency signals and unaffected by the high frequency signals. The lower capacitance beyond either side of the specific voltage of 0.8 V is owing to the dominating behavior of the effective biasing of the junction in both forward and reverse directions. At the lower voltages, the junction capacitance dominates, and at higher applied bias voltage,

the diffusion capacitance dominates [51].

Moreover, the frequency-dependent C–V characteristics are drawn in Fig. 14, and the inset shows the enlarged portion of the effective region. For all frequencies, the capacitance is mostly independent of the applied bias voltage until 0.5 V. At the same time, the capacitance shows an exponential increase of about 0.6 V in the lower frequency region below 10^5 Hz . This effect is observed on account of voltage dependent lower frequency capacitance and the voltage independent higher frequency capacitance [52].

The illumination-dependent C–V characteristics in the range from 0.1 Sun to 2 Sun are plotted in Fig. 15. The capacitance of the cell depends on the photogenerated carriers, as explained in eq. 3. However, total capacitance consists of the photo capacitance and the depletion or geometrical capacitance. Further, the depletion capacitance varies exponentially with the applied bias voltage until the built-in voltage. Thus, in Fig. 15, we can see that the capacitance is mostly affected by the voltage, which fades away the effect of illumination. Thus, the capacitance shows an exponential rise with the applied bias voltage till about 0.6 V, which is in the forward region. On the other hand, after the peak has been attained, the capacitance falls off rapidly because of the breakdown, which increases the current in the diode region. Accordingly, the illumination-dependent C–V characteristics reveal that the proposed Si bottom cell will remain consistent in the tandem structure.

The MS characterization of the solar cell facilitates understanding the apparent doping concentration, built-in voltage [53] and defect concentration. The MS characterization of the Si bottom cell is accomplished under various frequencies from 10^{-2} Hz to 10^6 Hz , as displayed in Fig. 16. For all the frequencies, the MS plots show a slight variation over the applied bias voltages, as shown in the inset of Fig. 16. From Fig. 16, the built-in voltage may be estimated at approximately 0.6 V [54,55]. The MS plots show a slight decrement till the built-in voltage, thereafter the MS plots show an exponential rise with the applied bias voltage. This shifting behavior of the MS plot after the built-in voltage accounts for the diode behavior of the device [52].

In addition, the C–F characteristics of the Si bottom cell under various working temperatures are presented in Fig. 17. For all the temperatures, the capacitance is almost invariant till the frequency 10^5 Hz . Although at higher frequencies, the capacitance decreases slightly for all temperatures. This slight decrease in the capacitance can be ascribed to the inconsistency of high-frequency signals. At higher temperatures, the thermally generated charge carriers are enhanced, which further contributes to the capacitance of the cell.

Along with the temperature-dependent C–F characterization, the C–V characterization is also represented at various working temperatures. Fig. 18 reveals the C–V characterization at working temperatures in the range of 260 K and 350 K. It can be inferred from the concern that the capacitance increases with the applied bias voltage and attains a

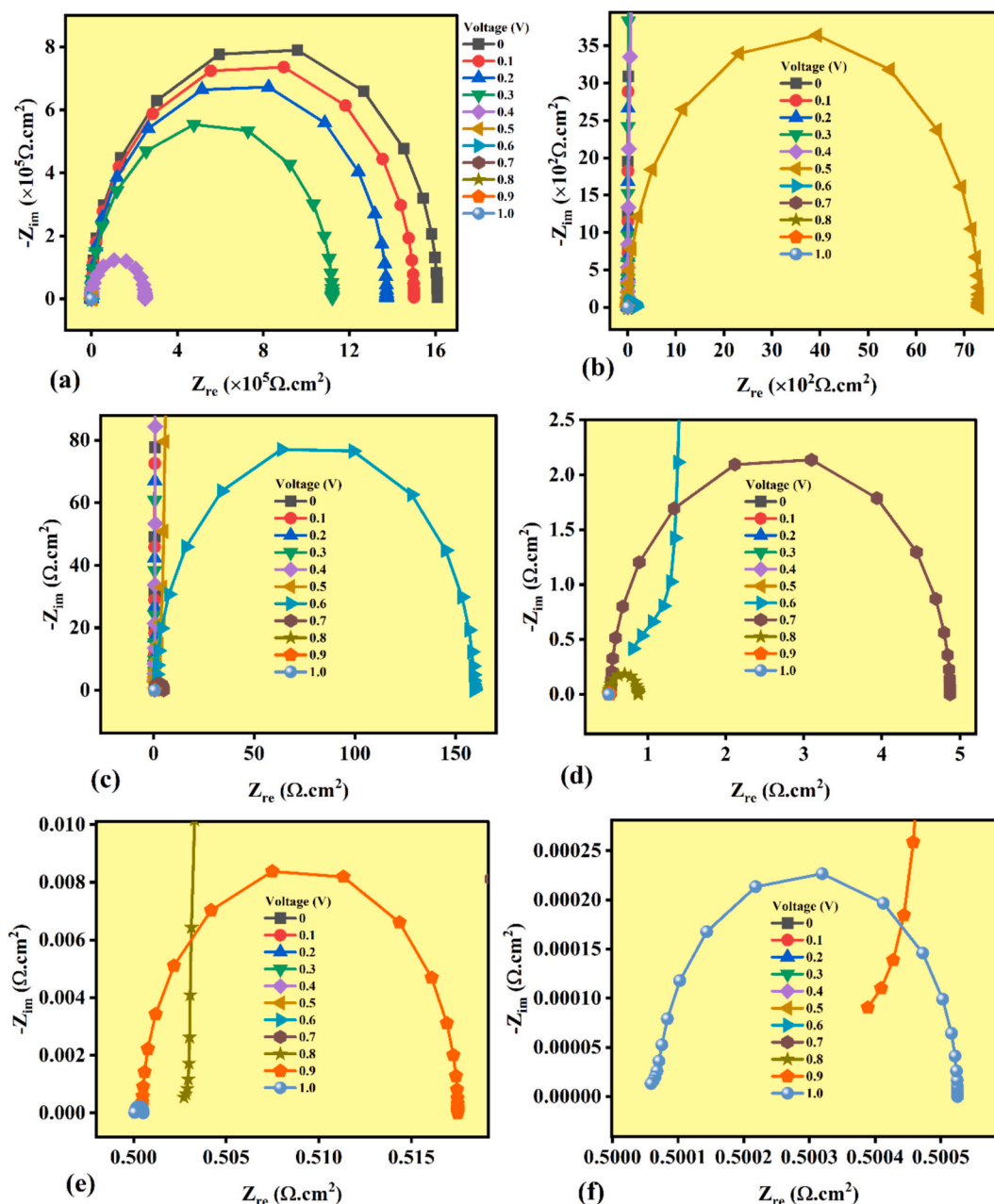


Fig. 21. (a) Voltage-dependent Nyquist plots for the Si-bottom cell, and (b) to (f) are the magnified scale graphs for precise variation.

peak of approximately 80 nFcm^{-2} , subsequently falling rapidly. As the temperature rises, the peaks of the C—V characteristics shift toward the lower biasing. This shifting of the peak shows that the recombination probability of the thermally generated carriers is higher than photo-generated carriers. Thus, the complete recombination was achieved for slightly lower applied bias voltages [56].

The charge transportation in the solar cells plays an important role, which the nature of Nyquist plots can recognise. Accordingly, the Nyquist plots are presented under various illumination levels in Fig. 19 (a & b) to understand the dynamical behavior of charge carriers in the Si bottom cell. Under the lower illumination level at 0.1 Sun, the semicircle of the Nyquist plot has the largest radius. The largest semicircle provides information on the impedance of the Si bottom cell, which is highest of the order of $10^6 \Omega \text{cm}^2$. In the low-frequency segments of the Nyquist plot, semicircles have larger separations, whereas the high-frequency ones are merged. This manner of the Nyquist plots is the affirmation of the dependence of impedance on the low-frequency signal and the

independence on the high frequencies.

Further, the effect of the working temperature on the complex impedance of the solar cell is observed in the feasible temperature range of 260 K to 350 K, as presented in Fig. 20. The semicircle of the Nyquist plot is slightly magnified for higher temperatures. This enlarged nature of the semicircle is attributed to the increased conductivity of the cell due to the thermally generated carrier concentration [57]. However, this increase is observed in the low-frequency segments of the Nyquist plots, while in the high-frequency region, the impedance is almost insensitive to the temperature. Ultimately, these temperature-dependent impedance spectra reveal that the shunt resistance increases with temperature from approximately $1.4 \text{ M}\Omega$ to $1.7 \text{ M}\Omega$. At the same time, the series resistance remains at about 0.5Ω , which reveals less degradation of the proposed bottom cell in the feasible temperature range.

The Nyquist plots, i.e., the imaginary impedance vs real impedance, explore the charge transport phenomenon at the interface of the electrochemical devices. Thus, the effect of applied DC bias voltage on the

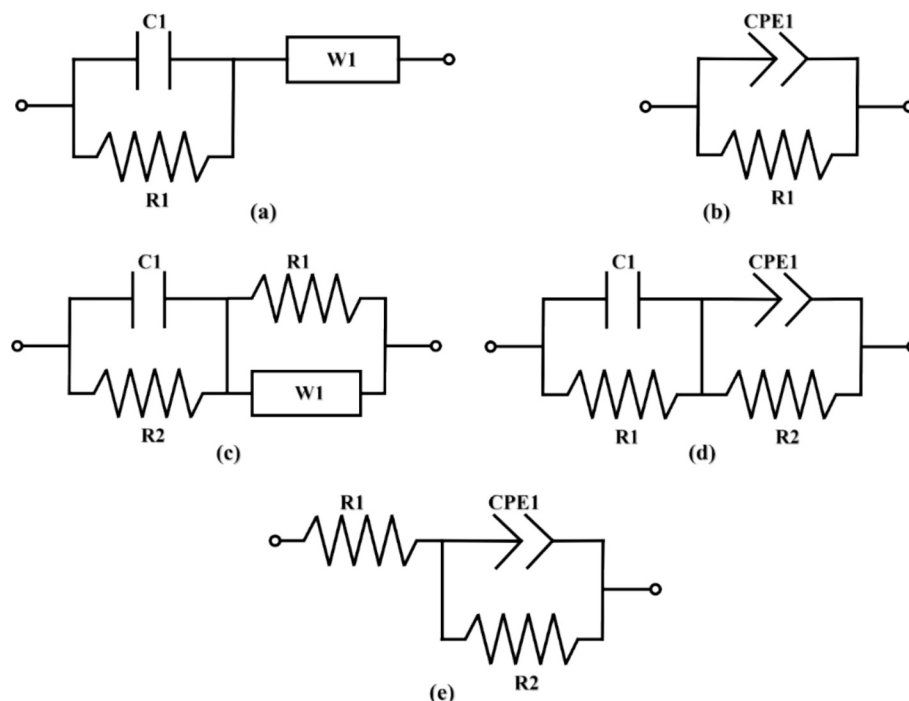


Fig. 22. Electrical equivalent circuits for the bottom cell at biasing voltages (a) 0.1 V, (b) 0.2 V, (c) 0.3, 0.4, 0.6, and 0.7 V, (d) 0.5 V, and (e) 0.8 V.

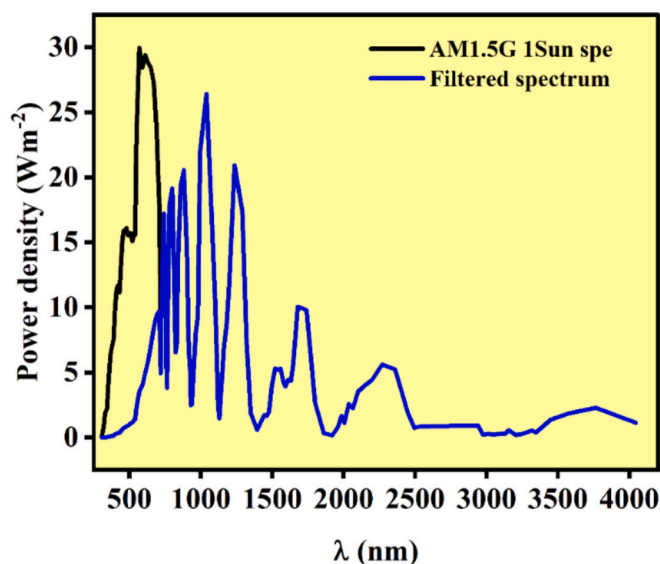


Fig. 23. Intensity (power density) of AM1.5G 1Sun used for the simulation of standalone cells and filtered spectrum used for the simulation of bottom cell.

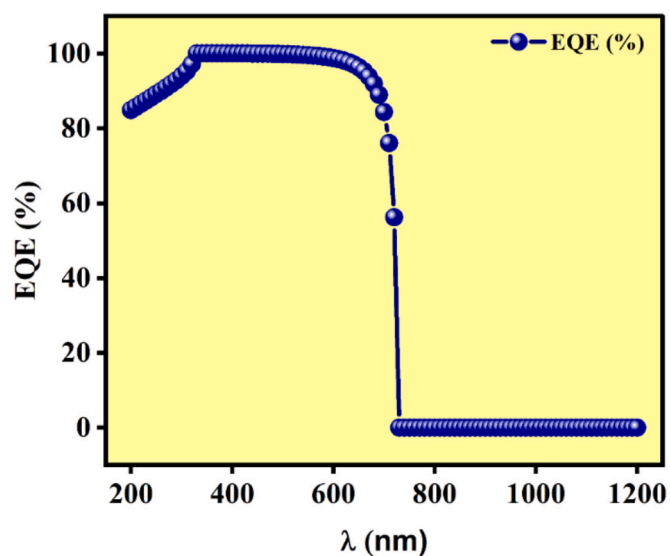


Fig. 24. EQE (External quantum efficiency) of the Sb_2S_3 top cell of the proposed tandem solar cell configuration.

Nyquist plots is examined for the bottom cell of the proposed tandem solar cell configuration as displayed in Fig. 21a. Figs. 21 b) to 21 f) shows the enlarged interpretation of the Nyquist spectra at various voltages from 0 to 1.0 V. The Nyquist plots show the shrinking nature of the semicircle with increasing applied bias voltage. The radii of the semicircles reduced in order from about $10^5 \Omega$ to $10^1 \Omega$ with the increase in the applied bias voltage. This reduction in the radii of the semicircles under the applied bias confirms increased recombination resistance at the interface of n-Si and p-Si layers, i.e., at the ETL/absorber interface of the bottom cell [58].

As far as experimental comparisons are concerned, the Nyquist plots we presented are in well agreement with an experimental study on Sb_2S_3 based solar cell with CdS buffer and TiO_2 ETL layers [59,60]. The

resistances (real and imaginary parts of impedance) obtained in our result are in the order of $10^3 \Omega \text{cm}^2$ that well followed the results in experimentally reported literatures. Further the built in voltage as observed in the Mott-Schottky plots, also matches closely with the experimental one [59,60]. Also, for the silicon bottom subcell the voltage dependent Nyquist plots show varying imaginary impedance in the order from $10^5 \Omega \text{cm}^2$ to $10^{-3} \Omega \text{cm}^2$ that corroborates with the experimental findings [61–63].

The EECs of the Si-bottom cell at various biasing voltages are drawn in Fig. 22. The components of the EECs have similar concerns, as discussed in Fig. 10 (EECs for the top cell). From Figs. 22 a) and 22 c), it can be observed that EECs consist Warburg element, which demonstrates the dominance of the diffusion process at voltages 0.1, 0.3, 0.4, 0.6, and 0.7

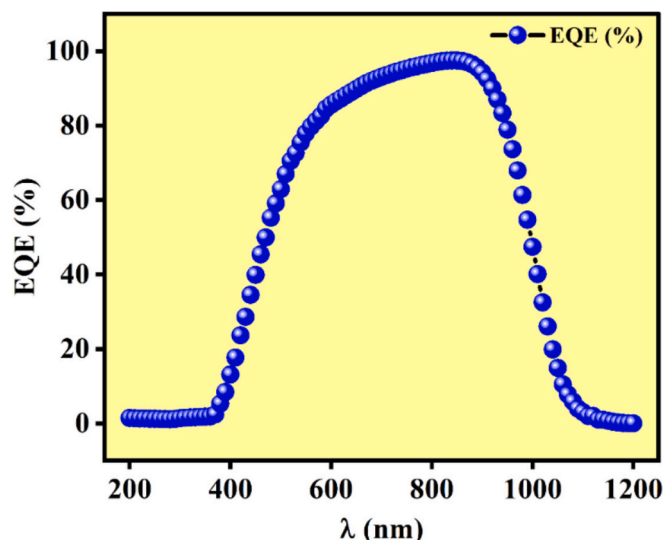


Fig. 25. EQE of the Si-bottom cell of the proposed tandem solar cell configuration.

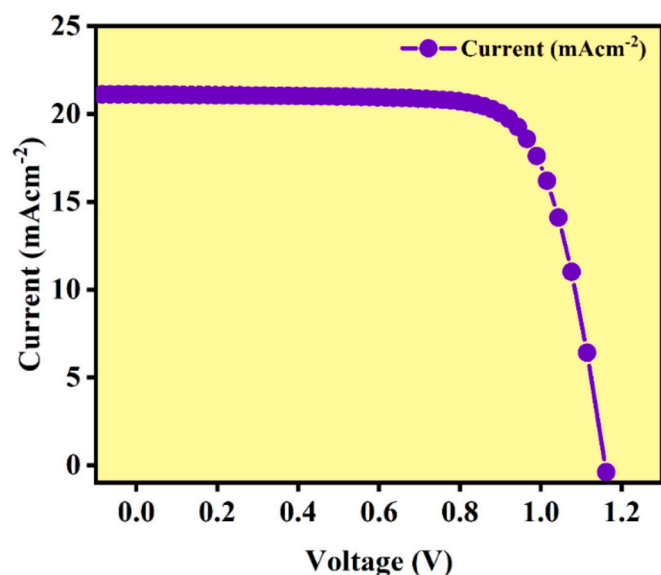


Fig. 26. I-V characteristics of the optimized and standalone top cell under standard working conditions and AM1.5G 1Sun illumination.

V. Further, the CPE elements present in the EECs, as shown in Figs. 22 b), 22 d), and 22 e), shows the non-ideal capacitive behavior of the Si-bottom cell for the biasing voltage levels 0.2, 0.5, and 0.8 V.

The tandem solar cell utilizes a stack of standalone solar cells, and they are ordered in a way that they absorb the solar spectrum depending on their bandgap. As there is a stack of solar cells, one by one, along the direction of exposure to sunlight. Thus, the solar cell, which is below the top cell, absorbs the reduced spectrum, which is the leftover after the absorption in the top cell. Therefore, to realize the tandem solar cell performance of the bottom cell, the filtered spectrum is applied during the simulation of the bottom cell. This filtered spectrum is estimated using the relation as mentioned in eq. 8. Fig. 23 shows the filtered spectrum as estimated using the relation as in eq. 8, and the AM1.5G 1Sun spectrum. The short-length photons near 500 nm absorbed by the top cell can be observed in Fig. 23.

The external quantum efficiency for the top cell and bottom cell is shown in Figs. 24 and 25. Since the quantum efficiency is the ratio of the

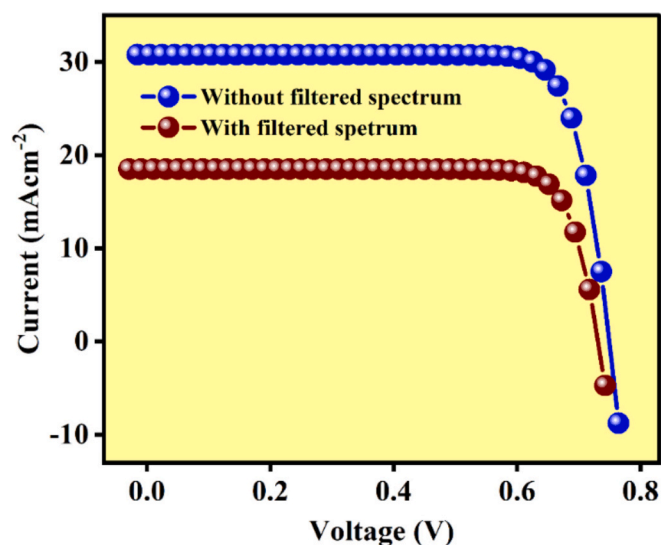


Fig. 27. I-V characteristics of the bottom cell with and without filtered spectrum (filtered spectrum estimated using eq. 8 for the top cell absorber thickness of 0.430 μm).

incident photon current to the photogenerated current of the solar cell. Moreover, the wide bandgap top cell absorbs short-wavelength photons of the incident solar spectrum of AM1.5G 1Sun (eq. 1). Thus, from Fig. 24, the maximum percentage of EQE for short wavelengths confirms the contribution of top cell performance to the performance of the tandem solar cell configuration.

The EQE of the Si-bottom cell is drawn in Fig. 25, which shows the maximum EQE value in the long wavelength range. Since the Si-bottom cell has a narrow bandgap therefore its absorption spectrum is in the range of long wavelengths (eq. 1). Thus EQE curve in Fig. 25 confirms the contribution of bottom cell performance to the performance of the proposed tandem solar cell configuration.

The finally optimized standalone top cell FTO/TiO₂/Sb₂S₃/V₂O₅ has been simulated at standard temperature 300 K under AM1.5G 1Sun spectrum for the I-V characteristics. The obtained I-V characteristic is drawn in Fig. 26.

Further, the I-V characteristic and corresponding performance parameters of the Si-bottom cell are obtained at the standard temperature condition of 300 K under AM1.5G 1Sun, as well as under the filtered spectrum (filtered spectrum estimated using eq. 8 for the top cell absorber thickness of 0.430 μm). The obtained I-V characteristics under AM1.5G 1Sun and under the filtered spectrum are plotted in Fig. 27. The I-V characteristic under the filtered spectrum corresponds to the contribution of the bottom cell to the tandem solar cell.

5. Conclusion

In this study, we present impedance spectroscopy analysis of both wide-bandgap and narrow-bandgap sub-cells intended for integration in a two-terminal (2 T) tandem solar cell architecture. Initially, a wide bandgap top cell with the configuration FTO/TiO₂/Sb₂S₃/V₂O₅ is simulated for the C-F characterization. The C-F characterization of the top cell revealed that the top cell has the lowest recombination for high-frequency signals. Further, the voltage-dependent C-F and frequency-dependent C-V characterization unveiled maximum recombination in the top cell near Voc. Secondly, the C-F and C-V characterization of the bottom cell with configuration n-Si/p-Si/p + -Si showed similar behavior to the top cell. Moreover, voltage and illumination-dependent MS analyses of both the cells disclosed the built-in voltages to be close to the Voc of the cells. Additionally, the Nyquist plots of the top and bottom cells unwrapped the EECs, with the electrical

components corresponding to the different processes in the device at various analogous frequency ranges. The voltage dependent EECs for the top cell suggests that the device shows non-ideal capacitive behaviour at higher (> 0.9 V) and lower (< 0.6 V) biasing voltages whereas at moderate voltages (between 0.6 V & 0.9 V) the device showed dominance of diffusion process. Similarly for the bottom cell, the voltages dependent EECs suggests for the device to have diffusion capacitive and non-ideal capacitive behaviours at lower and higher biasing levels respectively in the applied voltage range. Based on the impedance spectroscopic information of both the cells it is confirmed that they are suitable (as top and bottom cells) for the 2 T tandem configuration.

CRedit authorship contribution statement

Atish Kumar Sharma: Writing – original draft, Visualization, Validation, Software, Data curation. **Ritesh Kumar Chourasia:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nitesh K. Chourasia:** Writing – review & editing, Visualization, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Keyur Sangani:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization.

Ethical approval

Not required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] A. Richter, M. Hermle, S.W. Glunz, Reassessment of the limiting efficiency for crystalline silicon solar cells, *IEEE J. Photovolt.* 3 (2013) 1184–1191.
- [2] J. Chantana, Y. Kawano, T. Nishimura, A. Mavlonov, T. Minemoto, Thermodynamic limit of tandem solar cells under different solar spectra and their perovskite top solar cell, *Opt. Mater. (Amst.)* 113 (2021) 110819.
- [3] P. Kumar, A. Kumar, Double perovskite tandem solar cells: design and performance investigation of the use of CABB and CCSC as top and bottom cell absorber materials, *J. Electron. Mater.* 53 (2024) 2736–2752.
- [4] V.K. Singh, S. Srivastava, A.K. Singh, M.S. Chauhan, S.P. Patel, R.S. Singh, Theoretical study of highly efficient all-inorganic Sb₂S₃-on-Si monolithically integrated (2-T) and mechanically stacked (4-T) tandem solar cells using SCAPS-1D, *Environ. Sci. Pollut. Res.* 30 (2023) 98747–98759.
- [5] T. Ben Nasr, H. Maghraoui-Meherzi, H. Ben Abdallah, R. Bennaceur, Electronic structure and optical properties of Sb₂S₃ crystal, *Phys. B Condens. Matter* 406 (2011) 287–292.
- [6] M. Tamilselvan, A. Byregowda, C.-Y. Su, C.-J. Tseng, A.J. Bhattacharyya, Planar heterojunction solar cell employing a single-source precursor solution-processed Sb₂S₃ thin film as the light absorber, *ACS Omega* 4 (2019) 11380–11387.
- [7] R. Kondrotas, C. Chen, J. Tang, Sb₂S₃ solar cells, *Joule* 2 (2018) 857–878.
- [8] F. Caruso, M.R. Filip, F. Giustino, Excitons in one-dimensional van der Waals materials: $\{\text{Sb}\}_{2\text{S}_3}$ nanoribbons, *Phys. Rev. B* 92 (2015) 125134.
- [9] J. Zhou, H. Chen, X. Zhang, K. Chi, Y. Cai, Y. Cao, J. Pang, Substrate dependence on (Sb₄Se₆)_n ribbon orientations of antimony selenide thin films: morphology, carrier transport and photovoltaic performance, *J. Alloys Compd.* 862 (2021) 158703.
- [10] E.M. Mkawi, Solvothermal orthorhombic Sb₂S₃ nanobars: Effect of hydrothermal temperature in properties for solar cell application, *Respir. Physiol.* 19 (2020) 103603.
- [11] S. Mandati, N. Juneja, A. Katerski, A. Jegorov, R. Grzibovskis, A. Vembris, T. Dedova, N. Spalatu, A. Magomedov, S. Karazhanov, V. Getautis, M. Krunk, I. Oja Acik, 4.9% efficient Sb₂S₃ solar cells from semitransparent absorbers with Fluorene-based Thiophene-terminated hole conductors, *ACS Appl. Energy Mater.* 6 (2023) 3822–3833.
- [12] U.A. Shah, S. Chen, G.M.G. Khalaf, Z. Jin, H. Song, Wide bandgap Sb₂S₃ solar cells, *Adv. Funct. Mater.* 31 (2021) 2100265.
- [13] V.A. Tran, V.T. Le, T.T.T. Phan, M.T. Trinh, Emerging active materials for solar cells: Progress and prospects, *Adv. Electron. Mater.* 11 (2025) 2400530.
- [14] F. Priolo, T. Gregorkiewicz, M. Galli, T.F. Krauss, Silicon nanostructures for photonics and photovoltaics, *Nat. Nanotechnol.* 9 (2014) 19–32.
- [15] J.Y. Lee, J.E. Ju, C. Lee, S.M. Won, K.J. Yu, Novel fabrication techniques for ultra-thin silicon based flexible electronics, *Int. J. Extreme Manuf.* 6 (2024) 042005.
- [16] M. Boccard, L. Antognini, J. Cattin, J. Dr  n, W. Lin, V. Paratte, D. Turckay, C. Ballif, Loss analysis of a 24.4%-efficient front-junction silicon heterojunction solar cell and opportunity for localized contacts, *IEEE J. Photovolt.* 13 (2023) 663–671.
- [17] A. Blakers, N. Zin, K.R. McIntosh, K. Fong, High efficiency silicon solar cells, *Energy Procedia* 33 (2013) 1–10.
- [18] N. Liu, L. Wang, F. Xu, J. Wu, T. Song, Q. Chen, Recent Progress in developing monolithic perovskite/Si tandem solar cells, *Front. Chem.* 8 (2020) (2020).
- [19] Z. Ying, X. Yang, X. Wang, J. Ye, Towards the 10-year milestone of monolithic perovskite/silicon tandem solar cells, *Adv. Mater.* 36 (2024) 2311501.
- [20] Q. Wali, N.K. Elumalai, Y. Iqbal, A. Uddin, R. Jose, Tandem perovskite solar cells, *Renew. Sust. Energ. Rev.* 84 (2018) 89–110.
- [21] M. Okil, A. Shaker, I.S. Ahmed, T.M. Abdolkader, M.S. Salem, Design and analysis of Sb₂S₃/Si thin film tandem solar cell, *Sol. Energy Mater. Sol. Cells* 253 (2023) 112210.
- [22] Y. Zeng, J. Wu, K. Sun, J. Huang, L. Li, C. Sha, Y. Yao, Y. Lai, M. Green, F. Liu, X. Hao, Low-cost fabrication of Sb₂S₃ solar cells: direct evaporation from raw stibnite, *Ore Solar RRL* 6 (2022) 2100843.
- [23] J. Chen, G. Li, Z. Xu, C. Xu, F. Naveed, B. Liu, Y. Zhang, R. Zhou, C. Chen, M. Wang, J. Xu, L. Li, Recent advances and prospects of solution-processed efficient Sb₂S₃ solar cells, *Adv. Funct. Mater.* 34 (2024) 2313676.
- [24] I. Mora-Ser  , G. Garc  a-Belmonte, P.P. Boix, M.A. V  zquez, J. Bisquert, Impedance spectroscopy characterisation of highly efficient silicon solar cells under different light illumination intensities, *Energy Environ. Sci.* 2 (2009) 678–686.
- [25] E. Von Hauff, Impedance spectroscopy for emerging photovoltaics, *J. Phys. Chem. C* 123 (2019) 11329–11346.
- [26] M.M. Haque, S. Mahjabin, S. Khan, M.I. Hossain, G. Muhammad, M. Shahiduzzaman, K. Sopian, M. Akhtaruzzaman, Study on the interface defects of eco-friendly perovskite solar cells, *Sol. Energy* 247 (2022) 96–108.
- [27] Y. Cao, X. Zhu, J. Jiang, C. Liu, J. Zhou, J. Ni, J. Zhang, J. Pang, Rotational design of charge carrier transport layers for optimal antimony trisulfide solar cells and its integration in tandem devices, *Sol. Energy Mater. Sol. Cells* 206 (2020) 110279.
- [28] G. Li, F. Guo, X. Zhou, L. Xue, X. Huang, Y. Xiao, Design and simulation of Sb₂S₃ solar cells based on monolayer graphene as electron transport layer, *Opt. Mater. (Amst.)* 112 (2021) 110791.
- [29] Y. Etafa Tasisa, T. Kumar Sarma, R. Krishnaraj, S. Sarma, Band gap engineering of titanium dioxide (TiO₂) nanoparticles prepared via green route and its visible light driven for environmental remediation, *Results Chem.* 11 (2024) 101850.
- [30] S.M. Paraz, Jafri S.R. un N, Z. Tajvar, ul Hassan Alvi N, Q. Wahab, O. Nur, Effect of Annealing Atmosphere on the Diode Behaviour of ZnO/Si Heterojunction, *Elektronika ir Elektrotechnika* 27 (2021) 49–54.
- [31] S. Rahman, S.R. Al Ahmed, Photovoltaic performance enhancement in CdTe thin-film heterojunction solar cell with Sb₂S₃ as hole transport layer, *Sol. Energy* 230 (2021) 605–617.
- [32] K. Schneider, Optical properties and electronic structure of V₂O₅, V₂O₃ and VO₂, *J. Mater. Sci. Mater. Electron.* 13 (31) (2020) 10478–10488.
- [33] T.I. Alanazi, R.A. Alenazi, A.M. El Sayed, Tuning the band gap, optical, mechanical, and electrical features of a bio-blend for Cr₂O₃/V₂O₅ nanofillers for optoelectronics and energy applications, *Sci. Rep.* 1 (14) (2024) 12537.
- [34] H. Sun, Y. Zhou, Y. Xin, K. Deng, L. Meng, J. Xiong, L. Li, Composition and energy band-modified commercial FTO substrate for in situ formed highly efficient electron transport layer in planar perovskite solar cells, *Adv. Funct. Mater.* 29 (2019) 1808667.
- [35] S. Dabbabi, A. Garcia-Loureiro, M. Ajili, T. Ben Nasr, N. Kamoun, Enhanced power conversion efficiency of dye synthesized solar cell by few layered graphene/CuO nanocomposite as a working electrode dig, *J. Nanomater. Biostructures.* 12 (2017) 67–72.
- [36] S. Bhattacharya, S. John, Beyond 30% conversion efficiency in silicon solar cells: a numerical demonstration, *Sci. Rep.* 1 (9) (2019) 12482.
- [37] L.C. Andreani, A. Bozzola, P. Kowalczyk, M. Liscidini, L. Redorici, Silicon solar cells: toward the efficiency limits, *Adv. Phys. X* (2019) 4.
- [38] M. Burgelman, K. Decock, A. Niemegeers, J. Verschraegen, S. Degraeve, SCAPS Manual, 2025.
- [39] K. Kadan-Jamal, M. Sophocleous, A. Jog, D. Desagani, O. Teig-Susscholz, J. Georgiou, A. Avni, Y. Shacham-Diamand, Electrical impedance spectroscopy of plant cells in aqueous biological buffer solutions and their modelling using a unified electrical equivalent circuit over a wide frequency range: 4Hz to 20 GHz, *Biosens. Bioelectron.* 168 (2020) 112485.
- [40] D.K. Sarkar, A.K. Mahmud Hasan, M. Mottakin, V. Selvanathan, K. Sobayel, M. Ariful Islam, G. Muhammad, M. Aminuzzaman, M. Shahiduzzaman, K. Sopian, M. Akhtaruzzaman, Lead free efficient perovskite solar cell device optimization and defect study using Mg doped CuCrO₂ as HTL and WO₃ as ETL, *Sol. Energy* 243 (2022) 215–224.
- [41] I. Zonno, A. Martinez-Otero, J.C. Hebig, T. Kirchartz, Understanding mott-schottky measurements under illumination in organic bulk heterojunction solar cells, *Phys. Rev. Appl.* (2017) 7.
- [42] A.K. Sharma, A. Srivastava, P.K. Jha, K. Sangani, N.K. Chourasia, R.K. Chourasia, Signal components and impedance spectroscopy of potential p-Si/n-CdS/ALD-ZnO solar cells: EIS and SCAPS-1D treatments, *Adv. Theory Simul.* 7 (2024) 2400688.
- [43] N. Mondol, K.S. Islam, M.R. Islam, S.K. Das, Enhancing solar cell efficiency: investigation of high-performance lead-based perovskite-on-silicon (PVK-Si) tandem solar cells through design and numerical analysis, *AIP Adv.* 14 (2024) 015110.

- [44] X. Chen, X. Shu, J. Zhou, L. Wan, P. Xiao, Y. Fu, J. Ye, Y.T. Huang, B. Yan, D. Xue, T. Chen, J. Chen, R.L.Z. Hoye, R. Zhou, Additive engineering for Sb₂S₃ indoor photovoltaics with efficiency exceeding 17%, *Light Sci. Appl.* 1 (13) (2024) 281.
- [45] S.H. Im, H.J. Kim, J.H. Rhee, C.S. Lim, Seok S. Il, Performance improvement of Sb₂S₃-sensitized solar cell by introducing hole buffer layer in cobalt complex electrolyte, *Energy Environ. Sci.* 4 (2011) 2799–2802.
- [46] R. Wang, Y. Wang, Y. Pan, D. Qin, G. Weng, X. Hu, J. Tao, X. Luo, S. Chen, Z. Zhu, J. Chu, H. Akiyama, Improving the performance of Sb₂S₃ thin-film solar cells by optimization of VTD source-substrate proximity, *Sol. Energy* 220 (2021) 942–948.
- [47] W. Wang, J. Boneberg, L. Schmidt-Mende, Performance enhancement in Sb₂S₃ solar cell processed with direct laser interference patterning, *Sol. Energy Mater. Sol. Cells* 230 (2021) 111235.
- [48] A.S. Bondarenko, G.A. Ragoisha, *Progress in Chemometrics Research*, 2005, pp. 89–102.
- [49] N. Shibayama, Y. Zhang, T. Satake, M. Sugiyama, Modelling of an equivalent circuit for Cu₂ZnSnS₄- and Cu₂ZnSnSe₄-based thin film solar cells, *RSC Adv.* 7 (2017) 25347–25352.
- [50] V. Raman, R. Dongjoon, S. Aravindha Raja, K. Jihyun, P. Kandasamy, K. Joohoon, H.-K. Kim, High-performance flexible transparent micro-supercapacitors from nanocomposite electrodes encapsulated with solution processed MoS₂ nanosheets, *Sci. Technol. Adv. Mater.* 22 (2021) 875–884.
- [51] J. Zaraket, J. Sidawi, M. Aillerie, C. Salame, Capacitance evolution of photovoltaic solar modules under the influence of electrical stress, *Energy Procedia* 74 (2015) 1466–1475.
- [52] J.F. Chen, N.C. Chen, J.S. Wang, P.Y. Wang, Voltage and frequency dependence of differential capacitance in relaxed In_{0.2}Ga_{0.8}As/GaAs Schottky diodes, *Japanese J. Appl. Phys., Part 1 Regular Papers and Short Notes and Review Papers* 39 (2000) 1102–1103.
- [53] N.K. Sinha, D.S. Ghosh, A. Khare, Role of built-in potential over ETL/perovskite interface on the performance of HTL-free perovskite solar cells, *Opt. Mater. (Amst)*. 129 (2022) 112517.
- [54] O. Almora, C. Aranda, E. Mas-Marzá, G. Garcia-Belmonte, On mott-schottky analysis interpretation of capacitance measurements in organometal perovskite solar cells, *Appl. Phys. Lett.* (2016) 109.
- [55] E. Gutierrez-Partida, M. Rusu, F. Zu, M. Raoufi, J. Diekmann, N. Tokmoldin, J. Warby, D. Menzel, F. Lang, S. Shah, S. Shoaee, L. Korte, T. Unold, N. Koch, T. Kirchartz, D. Neher, M. Stollerfoht, Toward understanding the built-in field in perovskite solar cells through layer-by-layer surface Photovoltage measurements, *ACS Appl. Mater. Interfaces* 17 (2025) 11176–11186.
- [56] I. Khatiri, T.-Y. Lin, T. Yashiro, M. Sugiyama, Temperature-dependent current–voltage and admittance spectroscopy analysis on cesium-treated Cu (In_{1-x}Ga_x)Se₂ solar cell before and after heat-light soaking and subsequent heat-soaking treatments, *Prog. Photovolt. Res. Appl.* 28 (2020) 1158–1166.
- [57] M.M. Makhlof, H. Khallaf, M.M. Shehata, Impedance spectroscopy and transport mechanism of molybdenum oxide thin films for silicon heterojunction solar cell application, *Appl. Phys. A Mater. Sci. Process.* 128 (2022) 1–13.
- [58] M. Haider, C. Zhen, T. Wu, G. Liu, H.-M. Cheng, Boosting efficiency and stability of perovskite solar cells with nickel phthalocyanine as a low-cost hole transporting layer material, *J. Mater. Sci. Technol.* 34 (2018) 1474–1480.
- [59] Z. Feng, S. Sun, Y. Sun, X. Liu, H. Liu, H. Liu, Efficient CdS buffered TiO₂ electronic transport layer for Sb₂S₃ solar cells by a facile spinning coating process, *Appl. Physics A* 128 (6) (2022) 479.
- [60] H. Deng, Y. Cheng, Z. Chen, X. Lin, J. Wu, Q. Zheng, C. Zhang, S. Cheng, Flexible substrate-structured Sb₂S₃ solar cells with Back Interface Selenization, *Adv. Funct. Mater.* 33 (2023) 2212627.
- [61] D.A. van Nijen, S. Naoom, M. Muttillio, P. Procel, M. Zeman, O. Isabella, P. Manganiello, Analyzing the PN junction impedance of crystalline silicon solar cells across varied illumination and temperature conditions, *Sol. Energy Mater. Sol. Cells* 279 (2025) 113255.
- [62] S. Prayogi, D. Ristiani, D. Darminto, Unravelling the electrochemical impedance spectroscopy of hydrogenated amorphous silicon cells for photovoltaics, *Mater. Renew. Sustain. Energy* 14 (2025) 17.
- [63] J. Panigrahi, R. Vandana Singh, N. Batra, J. Gope, M. Sharma, P. Pathi, S. K. Srivastava, C.M.S. Rauthan, P.K. Singh, Impedance spectroscopy of crystalline silicon solar cell: Observation of negative capacitance, *Sol. Energy* 136 (2016) 412–420.